

Presentation on
“Customer Journey Intelligence”

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology

in

Department of Multidisciplinary Studies (Digital Marketing)

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Presentation Outline

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Problem Statement & Motivation

Real-World Problem

Online shoppers leave a trail of behaviour — visits, product views, carts and purchases — long before they decide to buy.

Where Existing Approaches Fall Short

- Analytics answer one question at a time: click probability, purchase probability, or one product to recommend.
- Funnel dashboards and RFM segments target broadly but never build a time-ordered journey per customer.
- Tools rarely say why an action is suggested now, and almost never treat “do nothing” as a valid choice.

Motivation

ML is now central to adaptive marketing [1] and joining behavioural signals in one framework beats isolated rule-based models [2].

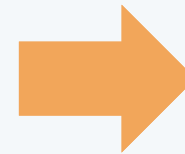
The Proposed Shift

- 1 Chronological, session-aware journeys
- 2 Journey stage & behavioural segments
- 3 Sequence-aware next-event & purchase prediction
- 4 Explainable Next Best Marketing Action, including “no action”

Traditional vs. Journey-Intelligence Approach

TRADITIONAL APPROACH

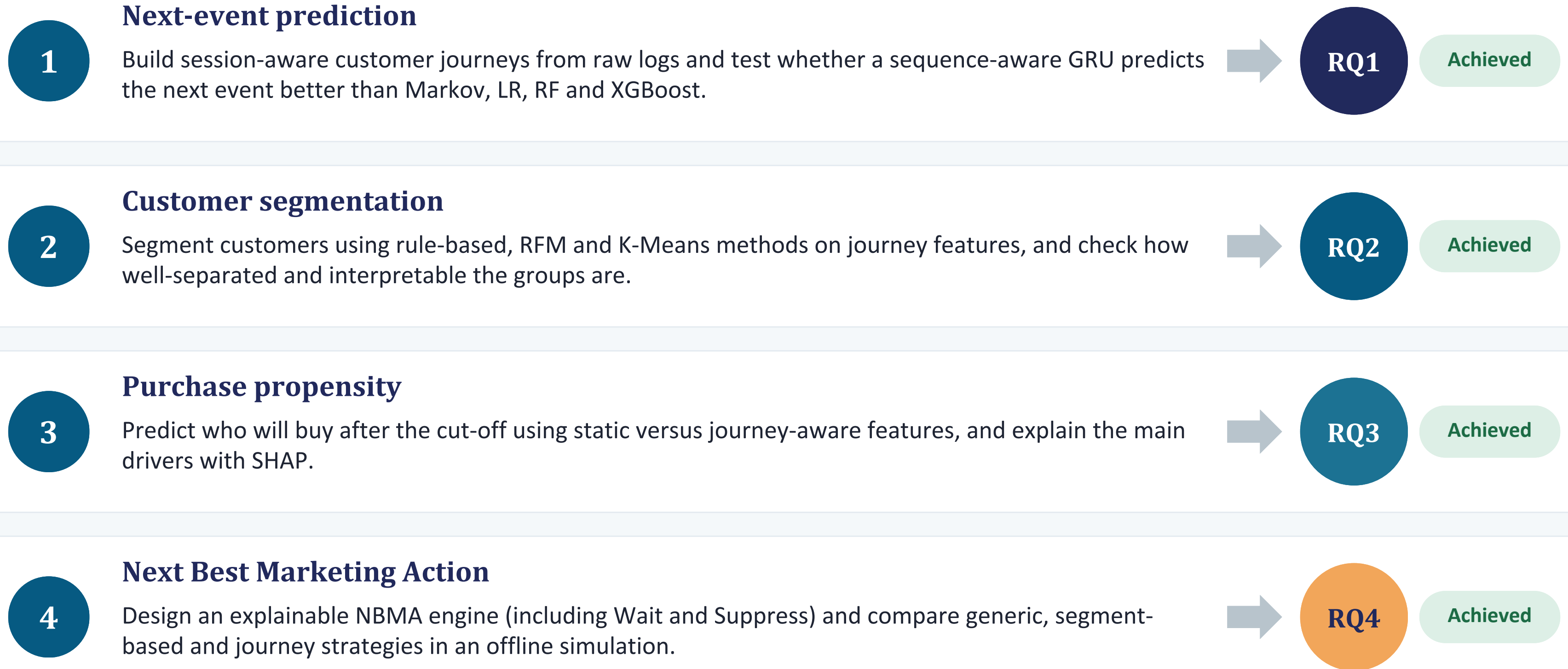
- Isolated-event models: CTR, purchase probability, single-product recommendation
- Static RFM / rule-based customer segments
- Big-data targeting rules applied broadly
- Funnel dashboards with no per-customer timeline
- No explanation for an action; “no action” rarely modelled



PROPOSED: JOURNEY INTELLIGENCE

- Session-aware journey per customer (30-min inactivity rule)
- Rule-based + RFM + K-Means segments built from journey features
- Next-event & propensity prediction: Markov → ML → GRU
- Journey-stage and funnel dashboards with drop-off and Customer 360°
- Explainable NBMA engine with Wait and Suppress as real actions

Objectives of the Project



Research Questions

RQ1

Does sequence-aware modelling (GRU) improve next-event prediction over a Markov baseline and classical ML?

RQ2

Does behavioural interaction data give separable, meaningful customer segments?

RQ3

Do journey-aware features improve purchase-propensity prediction over static features?

RQ4

Do journey-aware decision rules reduce irrelevant marketing actions versus generic and segment-based strategies?

Scope of the Project

1 Market & Competitor Analysis

Positioned against commercial NBA/CDP tools and recent AI-in-marketing literature.

Conceptual

2 Customer / Persona Analysis

Rule-based, RFM and K-Means segmentation of behavioural data.

Implemented

3 Website / Behavioural Analysis

Event standardisation, sessions, journey timelines and Customer 360° on RetailRocket.

Implemented

4 Marketing Strategy Design

Cart reminder, re-engagement, discount, cross-sell, wait and suppress via the NBMA engine.

Implemented

5 AI Integration

Markov, LR, RF, XGBoost and GRU models; K-Means; collaborative filtering; SHAP explanations.

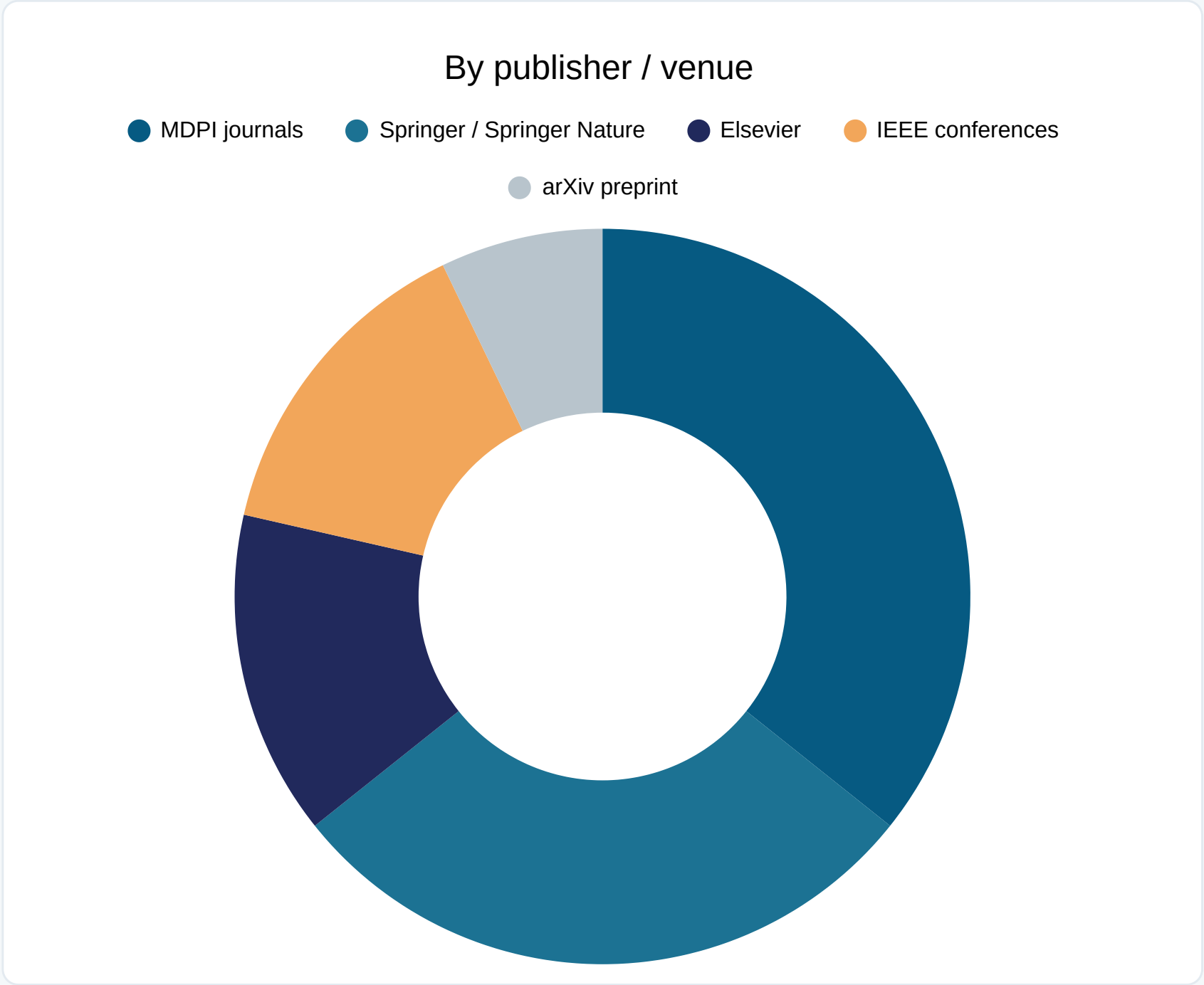
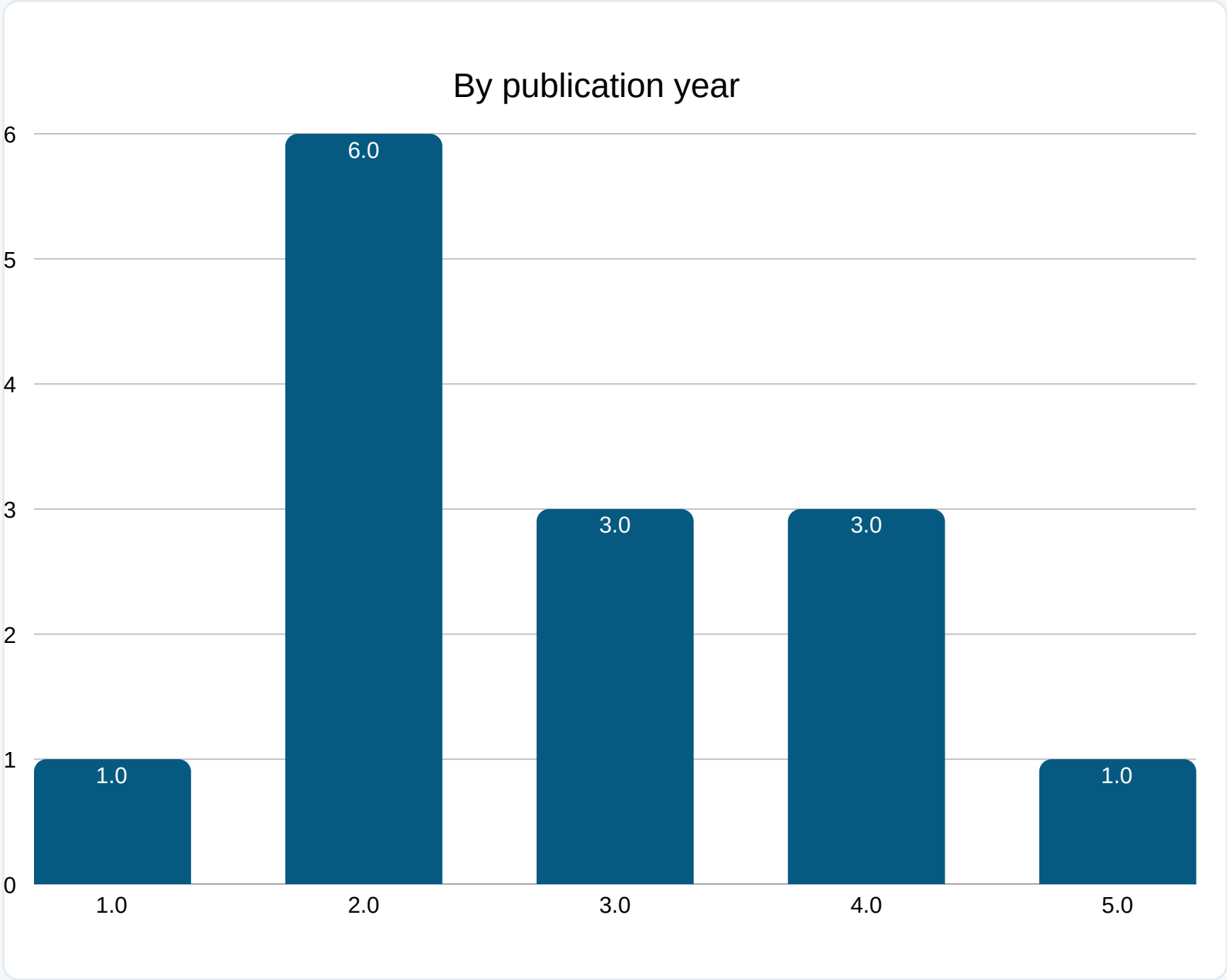
Implemented

6 Web Analytics & Evaluation

Executive dashboard, funnel and drop-off; offline evaluation of precision, coverage and actions.

Implemented (offline)

Literature Review — Bibliography Analysis



14 peer-reviewed / preprint sources (2020–2026) + 1 public dataset (RetailRocket, Kaggle) = 15 references

Literature Review Summary (1 of 2)

Ref.	Research Focus	Approach / Method	Key Finding	Gap This Project Addresses
[1]	AI in digital marketing	Comprehensive review	ML central to adaptive marketing	No per-customer decision layer
[2]	Customer journey optimisation	LDA + deep ensemble, channel allocation	~15% relative accuracy gain over classical ML	No explicit journey stages or no-action option
[3]	Customer segmentation	Review; RFM + K-Means	RFM + K-Means most used, not proven best	Static summaries, no chronological journey
[4]	Big-data strategic marketing	Dynamic-capabilities model	Big data supports adaptive decisions	Strategy level only; no per-customer engine
[5]	Explainable CRM → CJM	LightGBM + SHAP + QFD	Accuracy close to LSTM/CNN, faster training	Stops at explanation; no wait / suppress
[6]	Dynamic personalisation	Counterfactual analysis, K-Means, LR	Behaviour transitions sharpen personalisation	Not compared with other strategies
[7]	Segmentation in finance	K-Means, agglomerative, mean-shift	K-Means silhouette 0.62; A/B gain >100 bp	Domain-specific; no journey sequence

Literature Review Summary (2 of 2)

Ref.	Research Focus	Approach / Method	Key Finding	Gap This Project Addresses
[8]	Purchase prediction	KNN, SVM, RF, BPNN, XGBoost	XGBoost best: accuracy 0.976	Prediction only; no marketer action
[9]	Purchase intent (clickstream)	MBT-POP sliding windows	Best F1 0.903 at 2-day window	Session intent only
[10]	Shopper intent	LSTM on minimal browsing	LSTM beats hand-crafted features	No link from prediction to action
[11]	Next-touchpoint prediction	Process mining + boosted trees	43.1% next-touchpoint accuracy	RPA focus, no segment layer
[12]	Purchase journey	Neural ensemble + Shapley	AUROC 0.838; touchpoint impact	Attribution only; no stage rules
[13]	Imbalanced purchase data	12 ML pipelines + SMOTE	LightGBM F1 0.694, ROC-AUC 0.924	Milder imbalance (10.8%) than ours
[14]	AI personalisation & trust	Survey + SEM	Trust and usefulness drive engagement	No behavioural modelling

Research Gaps Identified

Existing Approach	Limitation Identified	How This Project Addresses It
RFM / clustering segmentation [3]	No chronological journey per customer	Session-aware Journey Builder feeding segmentation
Big-data strategic marketing [4]	No per-customer decision engine	Rule-based, per-customer NBMA engine
SHAP / QFD CRM–CJM frameworks [5]	Stops at analysis; no wait or suppress	Wait and Suppress modelled as valid actions
Counterfactual segmentation [6]	Not compared with other strategies	Offline comparison: generic vs segment vs journey (RQ4)
Purchase & intent prediction [8]–[10], [13]	Accuracy only; no marketer-facing action	Prediction feeds an explained NBMA with SHAP reasons

No reviewed system joins journey construction, stage/segment classification, sequence prediction and an explainable action layer with a no-action option.

Proposed Approach & Methodology

01



Research & dataset finalisation

Literature review, gap analysis, RetailRocket chosen

02



Data pipeline

Validation, de-duplication, 30-min sessionisation

03



Core backend

FastAPI services + PostgreSQL store

04



ML modelling

Segmentation, next-event and propensity models

05



Decision engine (NBMA)

Rules, personalisation, SHAP explanations

06



Frontend dashboards

React + Recharts: funnel, segments, Customer 360°

07



Strategy experiments

Offline simulation of four strategies (RQ4)

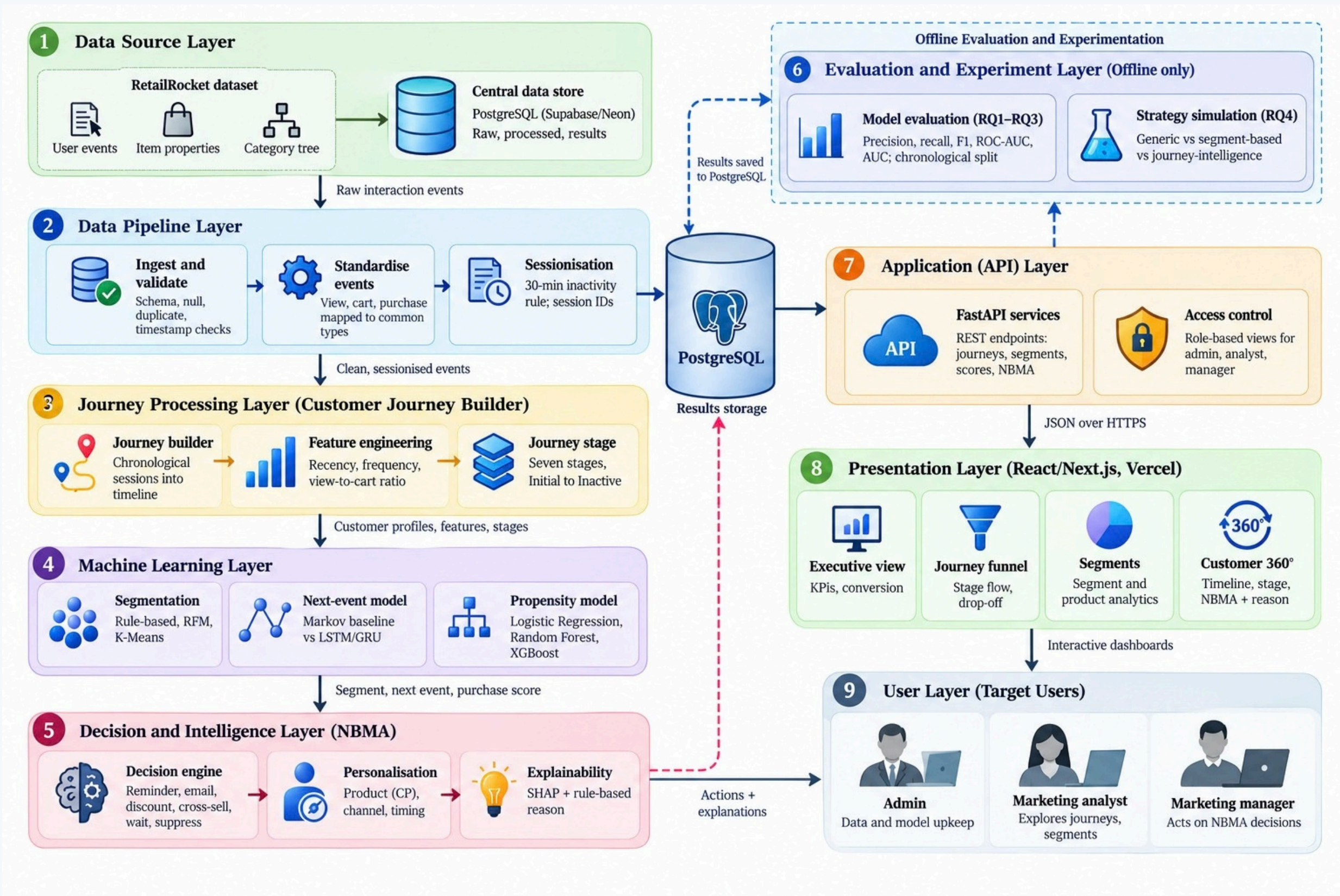
08



Documentation

IEEE research paper, report and reproducible scripts

System Architecture



Nine layers, top to bottom

- 1-2 Data source & pipeline: load, validate, sessionise
- 3 Journey builder: timeline, features, stage
- 4 ML: segments, next event, propensity
- 5 NBMA: action, personalisation, reason
- 6 Offline evaluation (RQ1-RQ4)
- 7 FastAPI + role-based access
- 8-9 React dashboards for admin, analyst, manager

Mathematical Model

Sessionisation

$$\text{new session} \Leftrightarrow t_{j+1} - t_j > \theta, \theta = 30 \text{ min}$$

Splits each visitor's events into sessions

Funnel ratios

$$\rho_{vc} = N_{cart} / (N_{view} + 1), \rho_{cp} = N_{buy} / (N_{cart} + 1)$$

View-to-cart and cart-to-purchase conversion

Segment quality

$$sil(u) = (b(u) - a(u)) / \max\{a(u), b(u)\}$$

Silhouette for choosing K (RQ2)

NBMA policy

$$a^*u = \pi(su, gu, \hat{e}u, \hat{p}u, ru)$$

Stage, segment, next event, propensity, time since buy

RFM features

$$Ru = \tau - tu^{\text{last}}, Fu = nu, Mu = \sum 1\{\text{purchase}\}$$

Recency, frequency, monetary proxy (items bought)

Markov baseline

$$P(b \mid a) = N(a \rightarrow b) / \sum_{b'} N(a \rightarrow b')$$

First-order next-event transition (RQ1)

Purchase propensity

$$\hat{p}u = P(yu = 1 \mid xu(\tau)), yu = 1\{\text{buy in } (\tau, \tau + H]\}$$

Label taken only after the cut-off τ (RQ3)

Strategy metrics

$$\text{Prectarget} = |T \cap B| / |T|, \text{Cov} = |T \cap B| / |B|$$

Targeting precision and coverage (RQ4)

Dataset Description

2,756,101

Total events

1,407,580

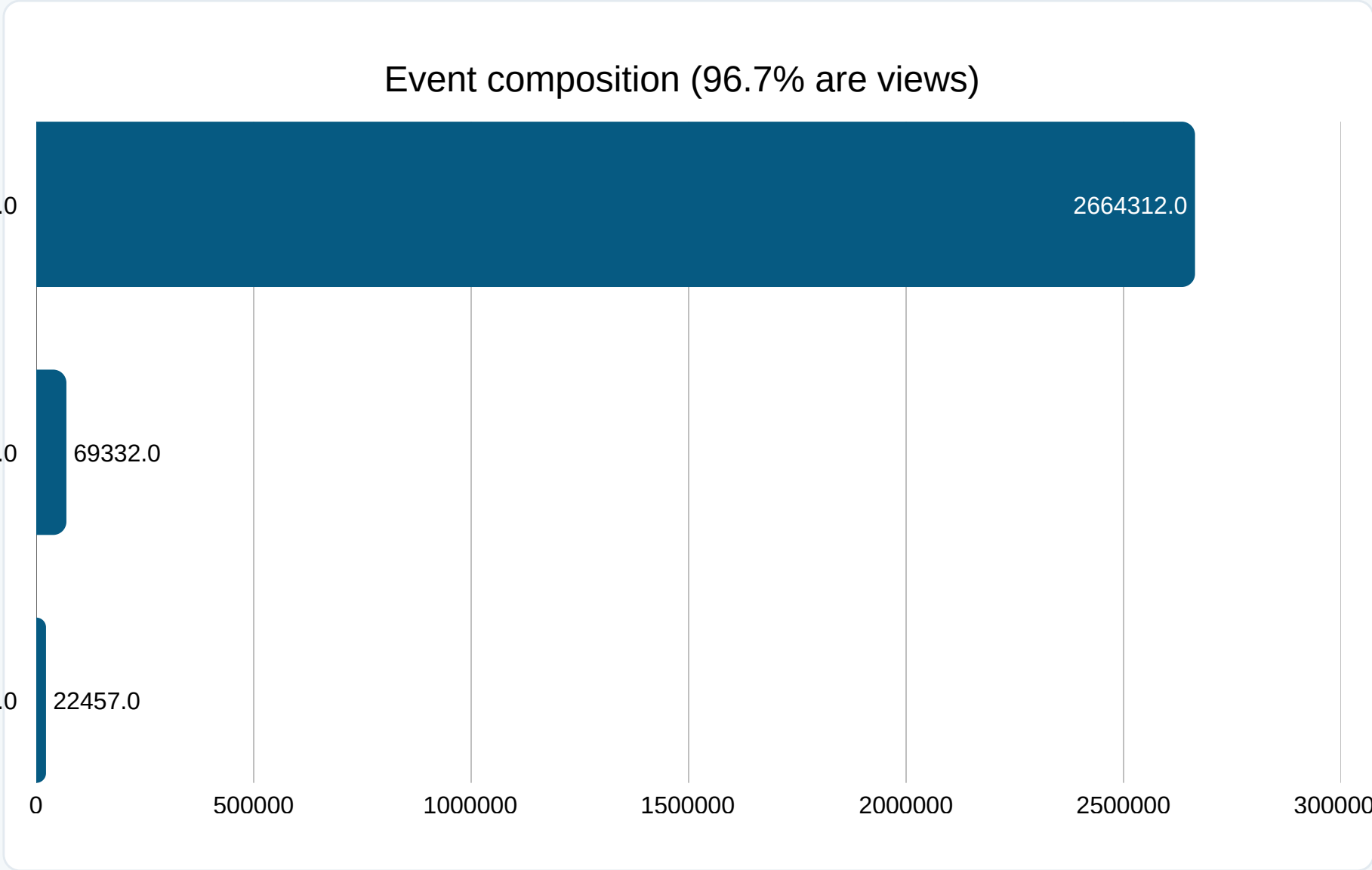
Unique visitors

417,053

Unique items

4.5 months

May–Sep 2015



Files & Structure

events.csv

visitorId, timestamp, event (view / addtocart / transaction), itemId, transactionId

item_properties.csv

20.2M timestamped rows: price, category, availability for 417K items

category_tree.csv

Category-to-parent map used for product analytics

Data Preparation & Experimental Setup

460

duplicates removed → 2,755,641 clean events

80%

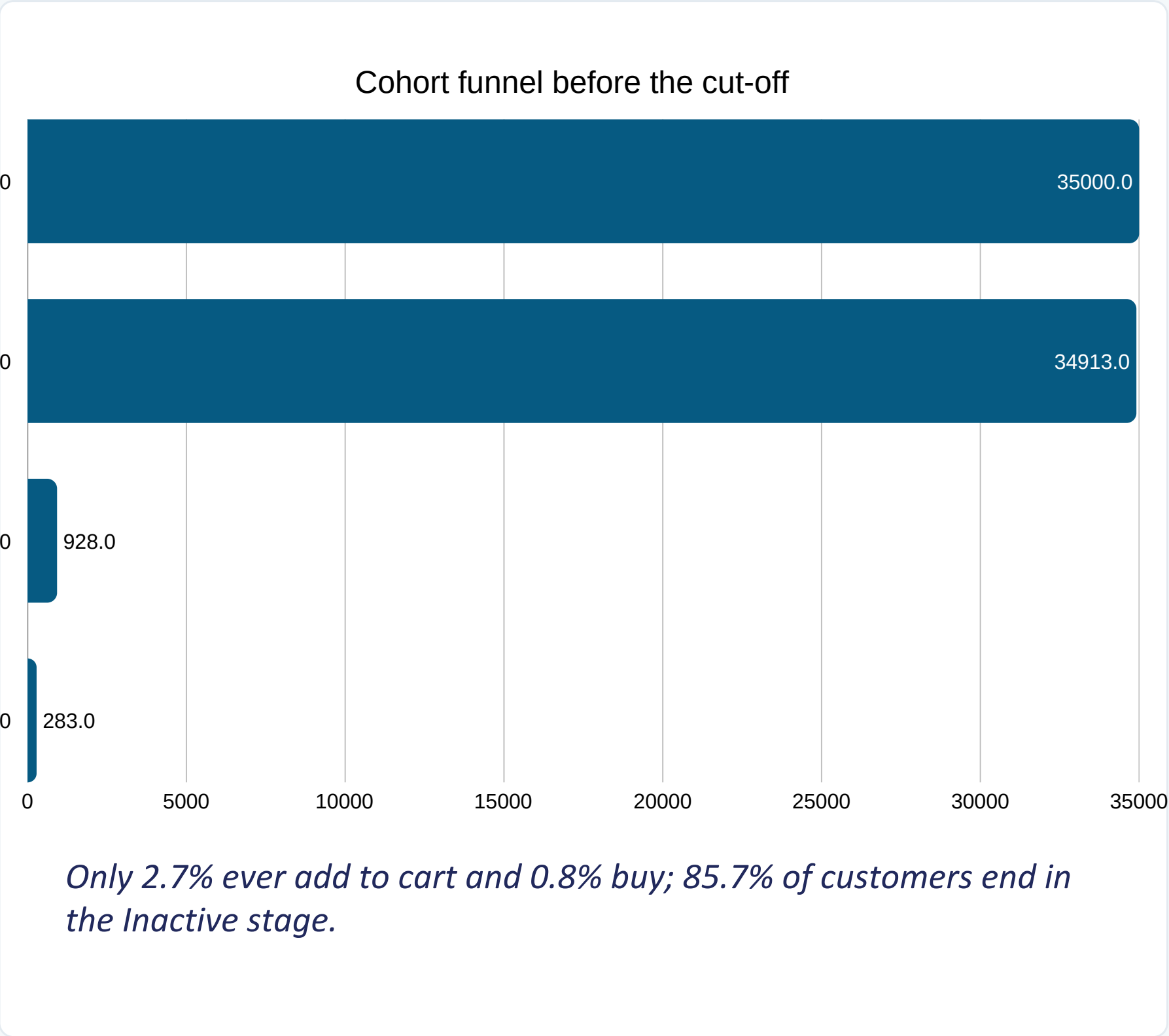
cut-off τ : features before, labels after (no leakage)

35,000

customer cohort: 24,500 train / 3,500 val / 7,000 test

0.07%

of the cohort buys after τ — an extreme class imbalance



Tools & Technologies



Frontend

React (Next.js), TypeScript, Tailwind CSS, shadcn/ui, Recharts



Backend

Python, FastAPI (REST: journeys, segments, scores, NBMA)



Database

PostgreSQL (Supabase / Neon): raw, processed and result tables



Machine Learning

Pandas, NumPy, Scikit-learn, XGBoost, PyTorch (GRU), SHAP



Visualisation

Recharts, Plotly, Matplotlib



Deployment

Vercel (frontend), Render / Railway (backend), Supabase / Neon

Journey Stages & NBMA Decision Engine

Seven journey stages (ordered, auditable rules)

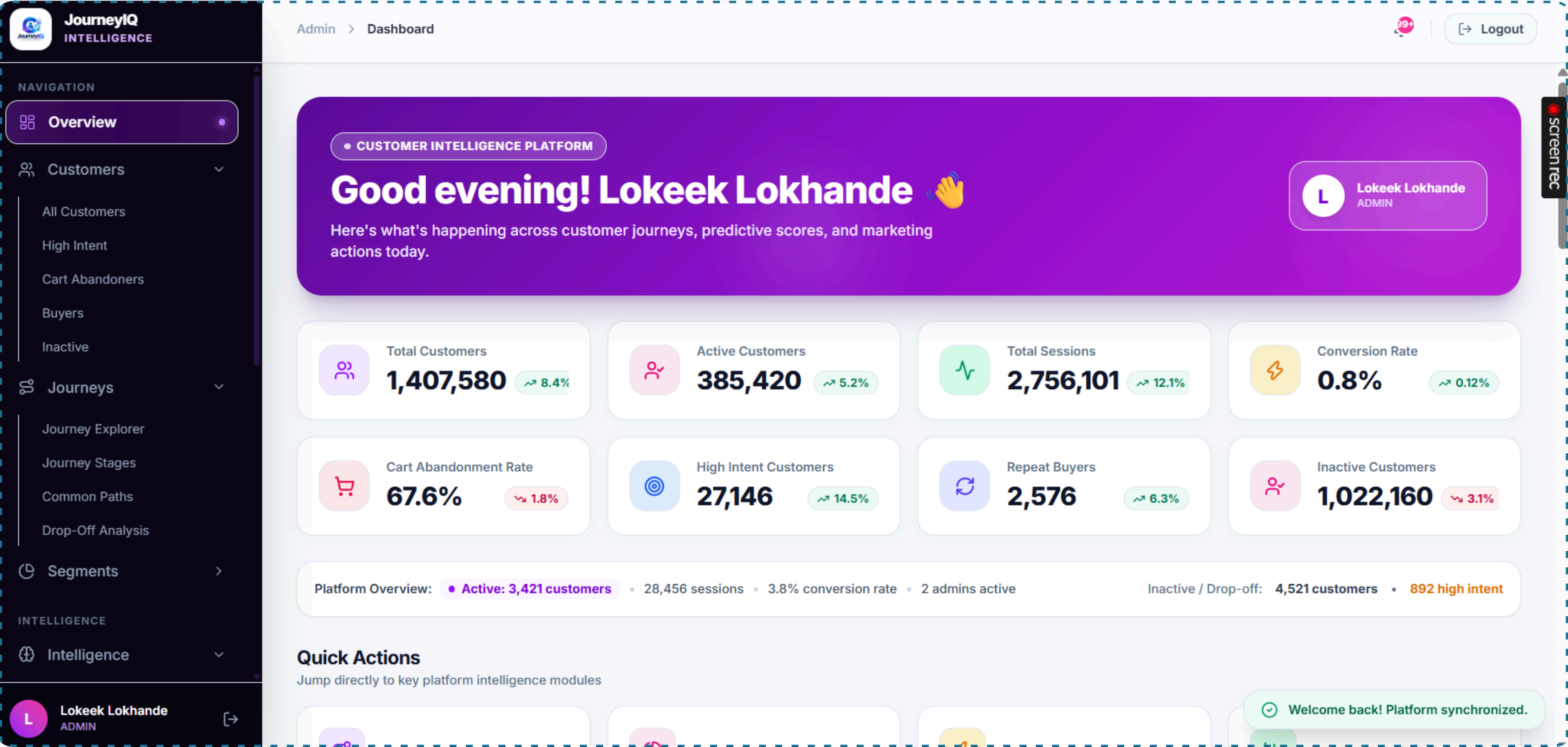


Priority	Condition	Action
1	Purchase within the last Δ days	Wait / Suppress
2	High Intent, cart present, no purchase	Cart Reminder
3	Consideration with high propensity $\hat{p}$	Targeted Discount
4	Retention segment, past purchase	Cross-Sell
5	Inactive but previously engaged	Re-engagement Email
6	Early stage, low propensity	Wait

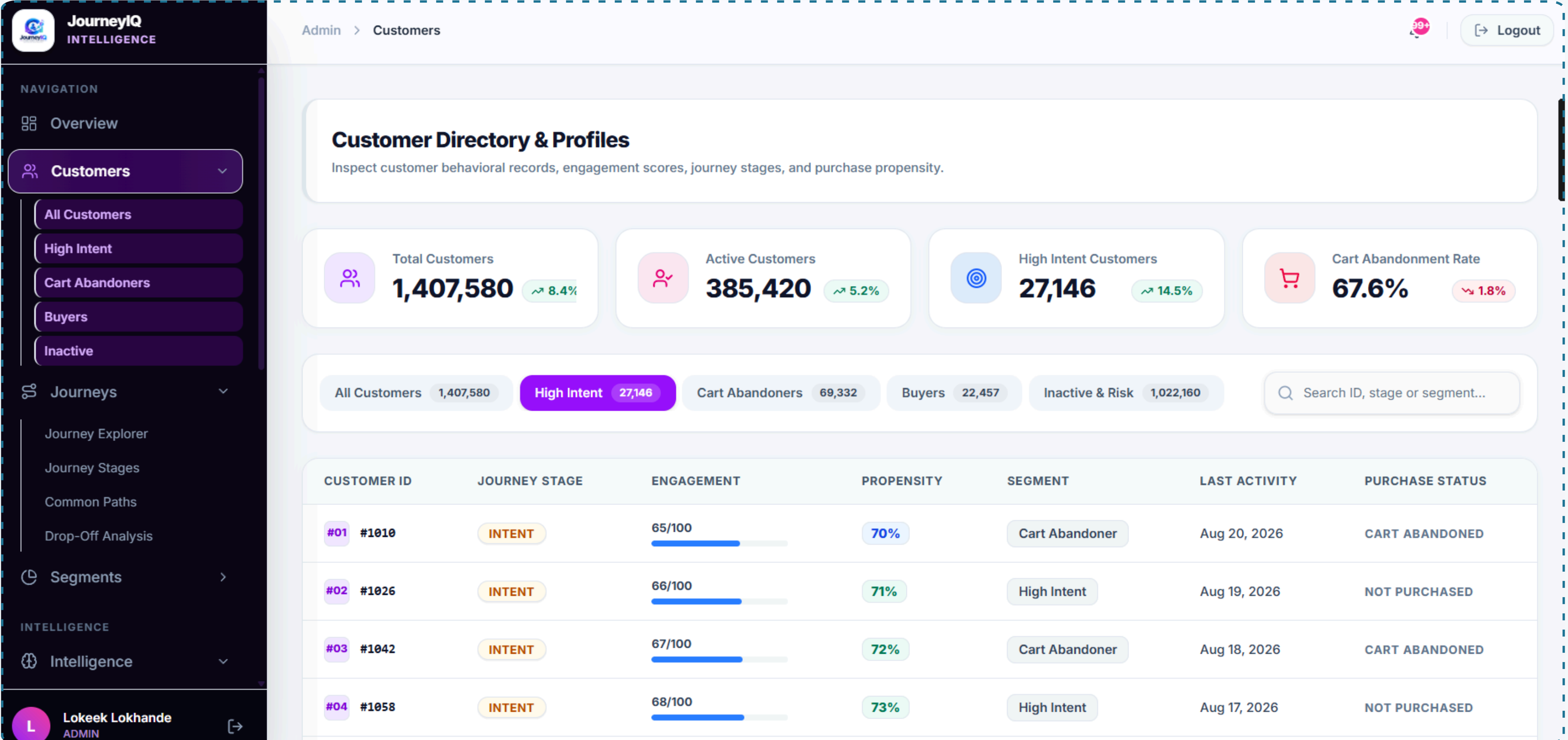
Every action comes with a reason

- Rule trace: which rule fired and the feature values it tested
- SHAP: top drivers of the purchase score
- Personalisation: product (item-based CF), channel and timing from activity pattern

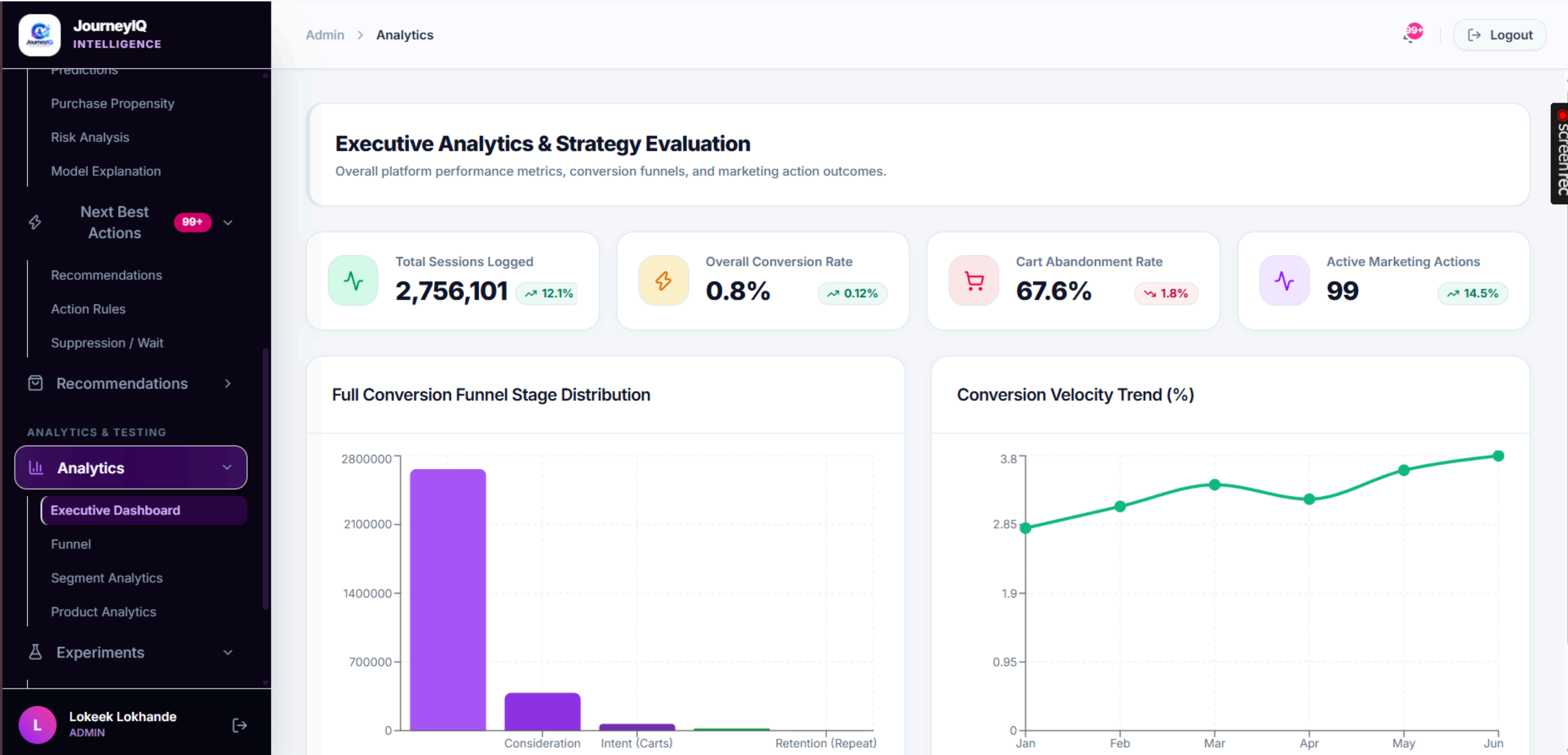
Implementation — Dashboards (1 of 3)



Implementation — Dashboards (1 of 3)



Implementation — Analytics (2 of 3)



Implementation — Models & Research (3 of 3)

JourneyIQ
INTELLIGENCE

Actions

- Recommendations
- Action Rules
- Suppression / Wait

Recommendations >

ANALYTICS & TESTING

- Analytics
- Executive Dashboard
- Funnel
- Segment Analytics
- Product Analytics

Experiments

- Strategy Comparison
- Model Evaluation

ADMINISTRATION

- Admin

L Lokeek Lokhande
ADMIN

Admin > Experiments > Strategies

99+ Logout

Marketing Strategy Comparison & Policy Simulation (RQ4)

Offline counterfactual evaluation comparing Generic Broadcast, Segment-Based Rules, and Journey-Intelligence Decision Policies on 7,000 held-out customer journeys.

Research Question 4 (Eq. 15 Protocol)

Evaluated Holdout Cohort

7,000

Journey Intelligence Suppr...

4,391

Marketing Noise Reduction

62.7%

Targeting Specificity (Hold...

0.2%

Policy Strategy Benchmark (Table XIII)
Empirical evaluation comparing baseline and journey-aware intervention models.

Journey Intelligence (Ours)Segment-BasedGeneric Blast

BASLINE 1Broadcast

Generic Marketing
Blasts uniform promotional messages to 100% of all arriving visitors.

Target Precision (Prectarget):0.26%

Audience Coverage:100% (7,000/7,000)

Unnecessary Interventions:71

Wait / Suppressed:0 (0.0%)

BASLINE 2RFM Rules

Segment-Based Strategy
Applies static segmentation rules (e.g., cart abandoners vs recent browsers).

Target Precision (Prectarget):0.26%

Audience Coverage:100% (7,000/7,000)

Unnecessary Interventions:285 (High)

Wait / Suppressed:0 (0.0%)

JOURNEY-AWAREProposed System

Journey-Intelligence
Multi-stage decision engine with explicit suppression and non-intrusive wait actions.

Target Precision (Prectarget):0.23%

Audience Coverage:33.3% (Targeted)

Unnecessary Interventions:149 (-47.7%)

Wait / Suppressed:4,391 (62.7%)

Results at a Glance



Next event

GRU

Macro-F1 0.465

vs ≈ 0.33 for all baselines

Supported



Customer groups

K-Means, K = 4

Silhouette 0.561

vs 0.553 for RFM K-Means

Supported



Purchase chance

Static LR

PR-AUC 0.148

journey features help ROC-AUC,
not PR-AUC

Partly supported



Strategy

Generic email-all

Prec 0.0026

journey rules hold back 4,391 /
7,000

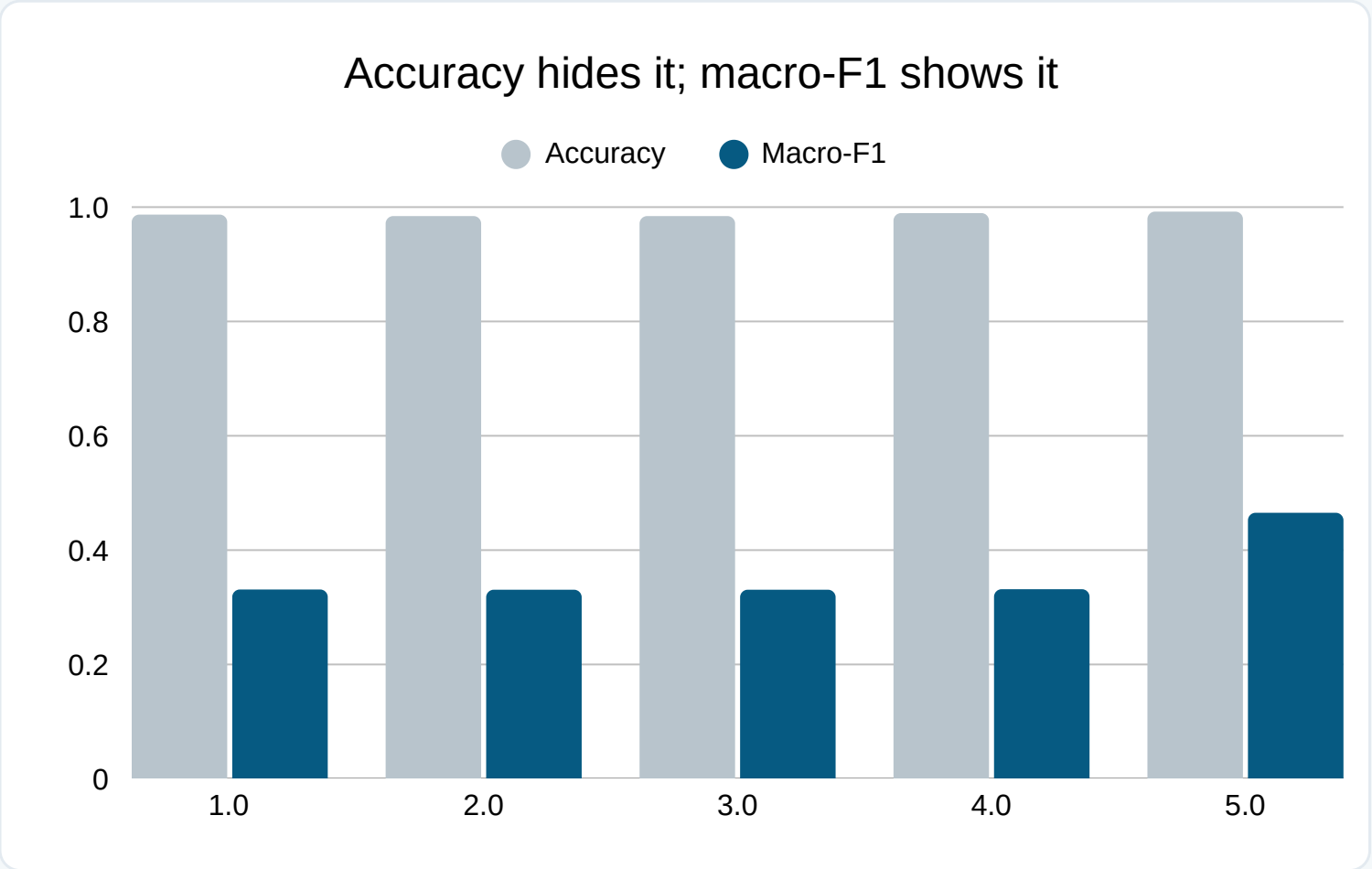
Not supported

Setup: cohort 35,000 · train 24,500 · val 3,500 · test 7,000 · RQ1 users 326 train / 382 test · buy-after-cut-off rate 0.07%

RQ1 Next-Event Prediction

Model	Accuracy	Precision	Recall	Macro-F1	Top-2
Markov chain	0.9869	0.3298	0.3325	0.3311	1.0000
Logistic Regression	0.9843	0.3298	0.3316	0.3307	0.9843
Random Forest	0.9843	0.3298	0.3316	0.3307	0.9843
XGBoost	0.9895	0.3298	0.3333	0.3316	0.9895
GRU (best)	0.9921	0.6640	0.4167	0.4653	1.0000

Precision, recall and F1 are macro-averaged over view, cart and purchase. Train 326 / test 382 users.

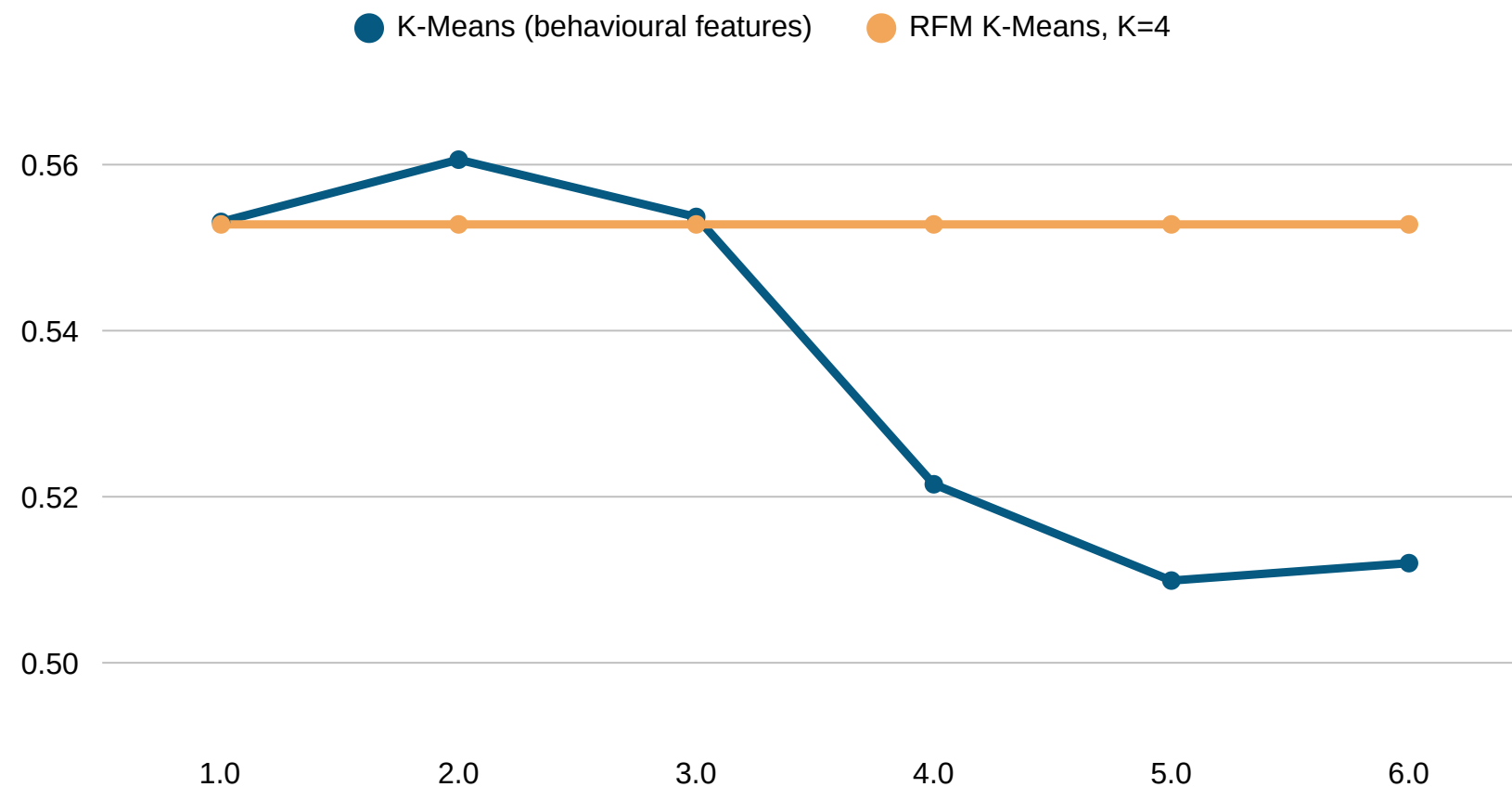


What it means

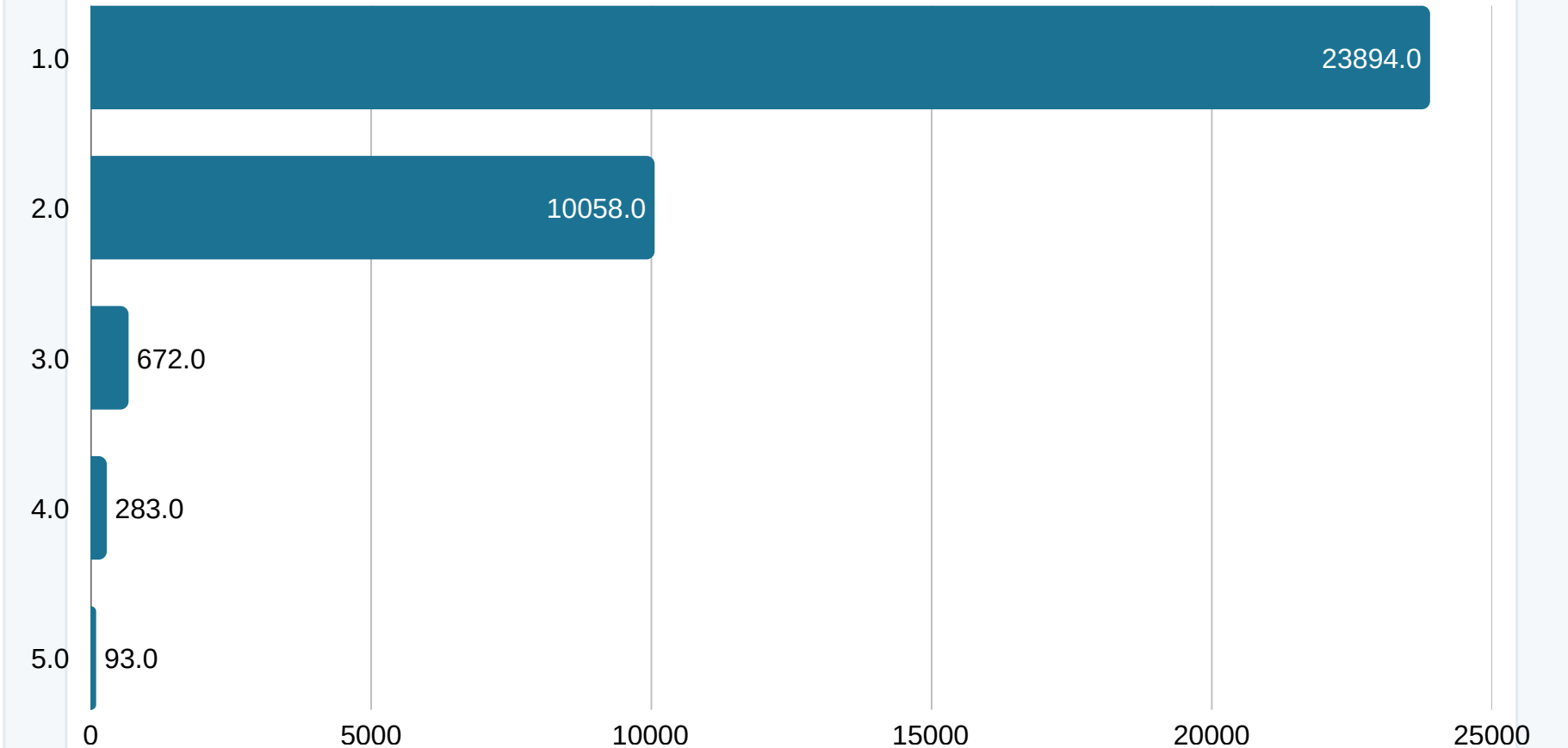
- **Accuracy is misleading:** almost every next event is a view, so always guessing “view” already scores ~98%.
- **Baselines stall at macro-F1 ≈ 0.33** — they only get the majority class right.
- **GRU is the only model that learns cart/purchase signals** (macro-F1 0.465, precision 0.664). RQ1 is supported, but 382 test users is small.

RQ2 Customer Segmentation

Silhouette score by number of clusters



Rule-based segment sizes

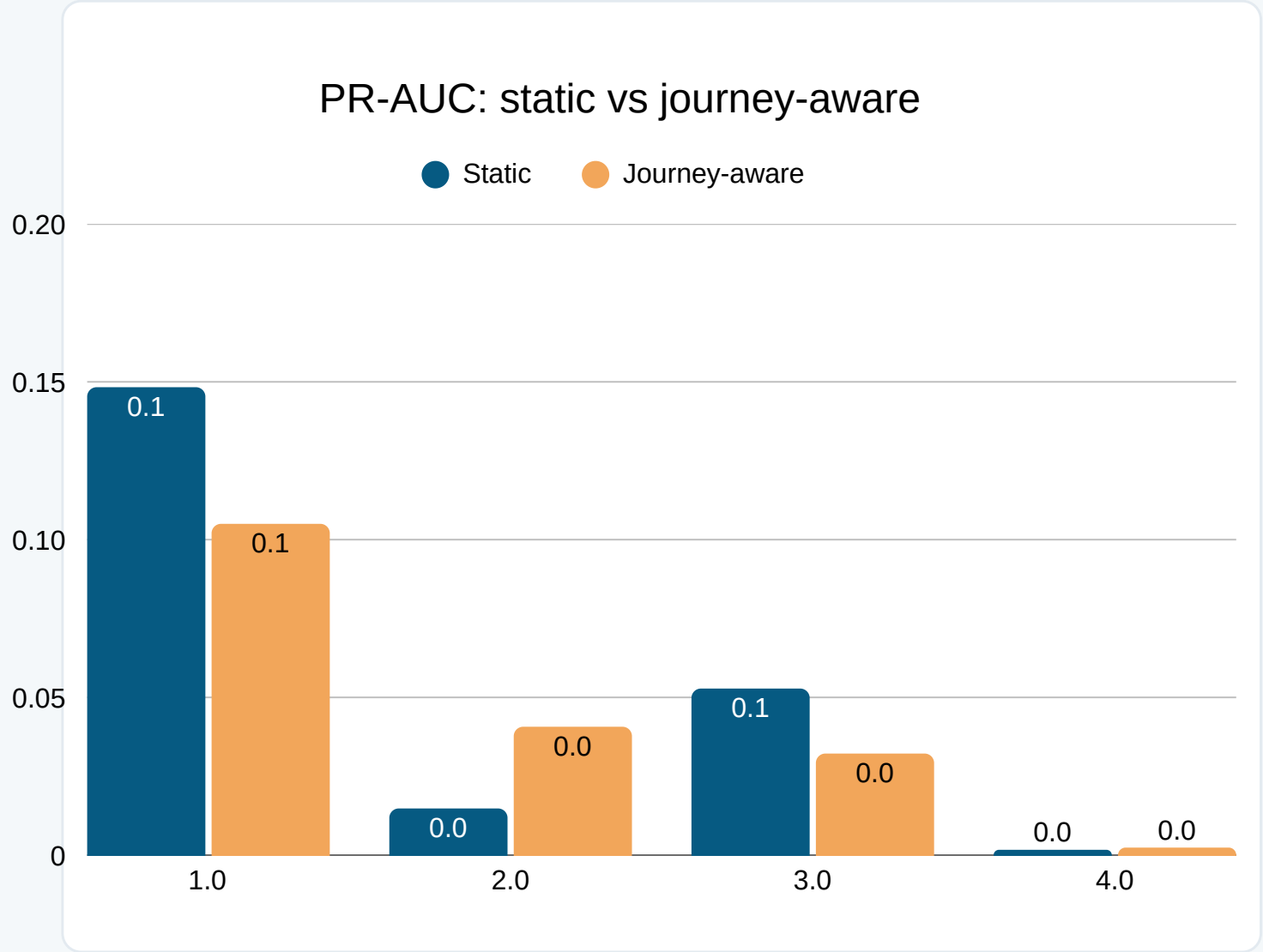


- **Best separation at K = 4 (silhouette 0.561);** K = 3 and 5 are close, then quality falls to ≈ 0.51 for $K \geq 6$.
- **Journey features add a small, steady gain** over RFM-only K-Means (0.553). Above 0.5 means compact, well-separated groups.
- **Groups are nameable but very uneven:** two-thirds are inactive, while the marketing value sits in a few small groups. RQ2 is supported.

RQ3 Purchase Propensity

Features	Model	Acc.	Prec.	Recall	F1	ROC-AUC	PR-AUC
Static	LR	0.8164	0.0055	0.3889	0.0108	0.5677	0.1483
Static	RF	0.9973	0.0000	0.0000	0.0000	0.6331	0.0148
Static	XGBoost	0.9974	0.0000	0.0000	0.0000	0.7473	0.0528
Static	GRU	0.3077	0.0008	0.2222	0.0016	0.1598	0.0017
Journey	LR	0.5274	0.0030	0.5556	0.0060	0.6274	0.1050
Journey	RF	0.9961	0.0000	0.0000	0.0000	0.7408	0.0407
Journey	XGBoost	0.9961	0.0000	0.0000	0.0000	0.7269	0.0322
Journey	GRU	0.9969	0.0000	0.0000	0.0000	0.3674	0.0024

Test n = 7,000. Ranked by PR-AUC because buyers are only 0.07% of the cohort.

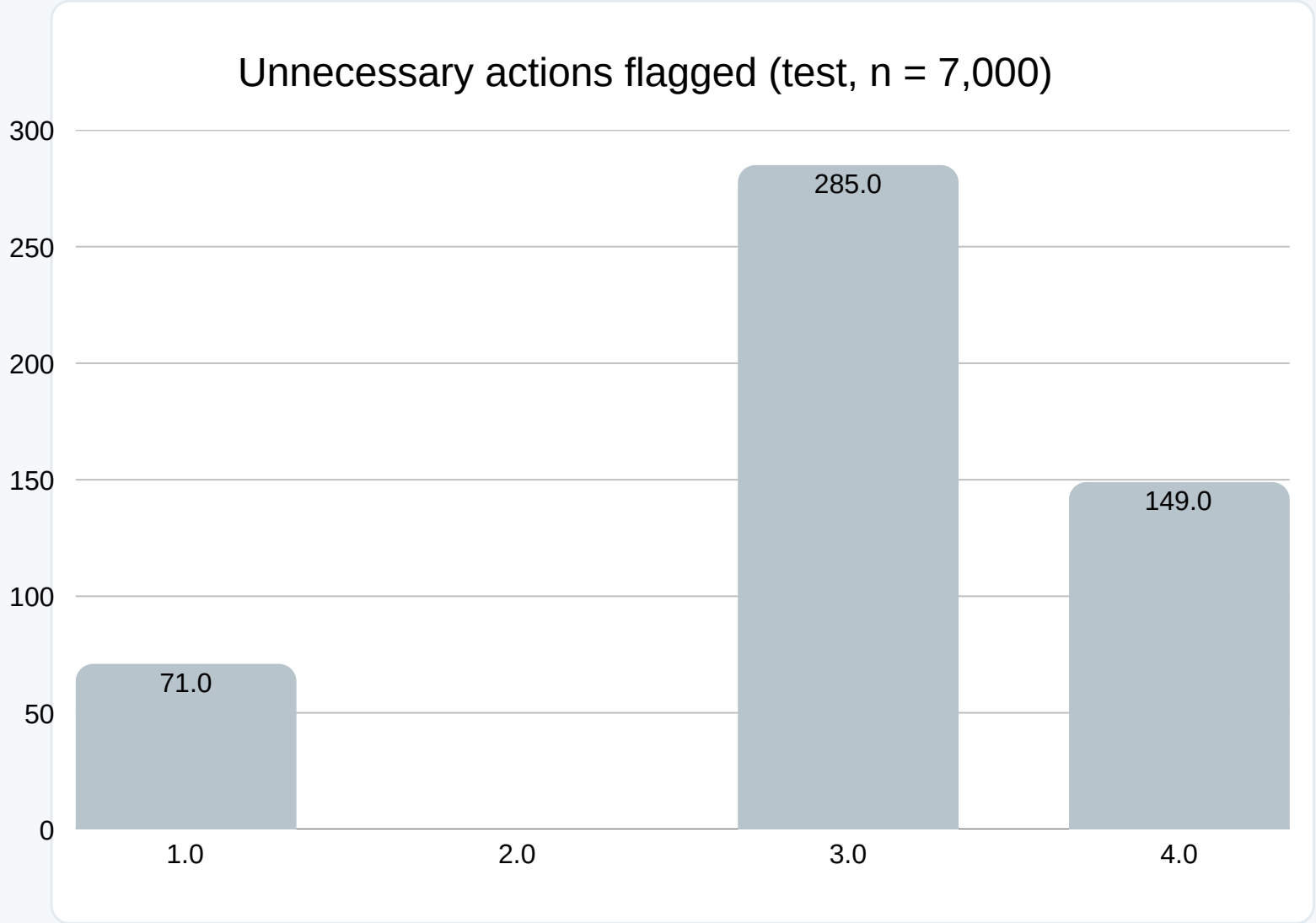


- **The 99.7% accuracy trap:** RF and XGBoost predict “no buy” for everyone and catch zero buyers. LR has lower accuracy but is the only useful model.
- **Journey features improve ranking:** ROC-AUC up for LR (0.57→0.63) and RF (0.63→0.74); LR recall up to 0.556. SHAP: stage_code 2.73, total_events 0.13, recency 0.03.
- **But static LR keeps the best PR-AUC (0.148),** so RQ3 is only partly supported.

RQ4 Offline Strategy Simulation

Strategy	Prec_target	Coverage	Unnecessary actions
Wait + Stop	0.0026	1.0000	71
Generic (email all)	0.0026	1.0000	0
Segment-based	0.0026	1.0000	285
Journey + rules	0.0023	0.3333	149

4,391 of 7,000 test customers get Wait or Stop under journey rules (≈ 63% fewer messages)



- **Generic email-all wins this simulation:** same top precision, full coverage, no flagged actions. RQ4 is not supported in this run.
- **Why:** offline, a message costs nothing; very few future buyers decide the metrics; NBMA thresholds were hand-set, not tuned.
- **Upside:** journey rules cut outreach by more than half at almost the same precision — worth testing once contact costs are counted.

Discussion



Accuracy is not the answer

With buys at 0.07%, a model that predicts “no” for everyone scores 99.9%. We judge RQ1 by macro-F1 and RQ3 by PR-AUC and recall.



Where journeys pay off

Sequence order helps next-event prediction (GRU) and journey features sharpen segments and ranking (ROC-AUC).



Where they don't yet

Rare purchases and hand-set thresholds limit propensity and strategy gains; email-all wins when messages are free.



Honest limits

Offline proxies, not causal uplift; one retailer, five months; 35K-customer cap; no channel-response data.

Research Ethics Principles



Honesty & Integrity

All results reported exactly as produced, including the ones that did not support our hypotheses (RQ3, RQ4).



No Fabrication / Falsification

No invented observations or tuned-to-look-good metrics; scripts regenerate every table.



No Plagiarism

All 14 papers, the RetailRocket dataset and open-source libraries are cited.



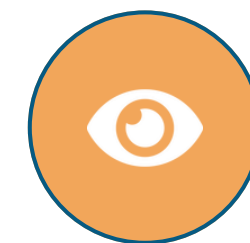
Data Privacy

Visitor and item IDs are pre-hashed; no personally identifiable information is used.



Fairness

Segments and NBMA rules reviewed so no group is unfairly over- or under-contacted.



Transparency & Responsible AI

Rule traces + SHAP make every action traceable; Wait and Suppress limit message fatigue.

Conclusion & Future Work

We built and evaluated an explainable Customer Journey Intelligence platform that turns raw clicks into journeys, stages and segments, predicts the next event and purchase chance, and recommends a Next Best Marketing Action — including a deliberate “no action”.

What we delivered

- 1 Full pipeline on 2.76M clean events with a leakage-free time split
- 2 GRU best for next event (macro-F1 0.465); K-Means K=4 best for segments (0.561)
- 3 Journey features improve ranking; static LR best PR-AUC (0.148)
- 4 Working web app: dashboards, Customer 360°, Models and Research pages
- 5 IEEE-format research paper on the framework and findings

Next steps

- Run on the full dataset instead of the 35K cohort to get more buyers
- Tune decision thresholds and add resampling (e.g. SMOTE) for RQ3
- Add a per-message cost to the RQ4 simulation
- Replace hand-set NBMA rules with uplift modelling
- Validate the policy with a live A/B test in a real store

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THANK YOU